

Anxiety & Anxiolytic Medications

Pharmacotherapy for anxiety disorders

Brain circuits in fear & anxiety

Transmitter systems implicated in anxiety

- CRF, NE & 5-HT, GABA

Anxiolytics & the treatment of anxiety

Fear and anxiety

Fear

- threat is temporally or spatially immanent
- requires the amygdala

Anxiety

- threat is temporally or spatially remote
- The extended amygdala maintains fear response

Survival

Both fear and anxiety help an organism survive encounters with environmental danger (threat).

The amygdala is critical for fear learning

Sensory input

- Thalamus
- Sensory cortex

Amygdala

- US and CS inputs converge in the amygdala
- output coordinates fear responses

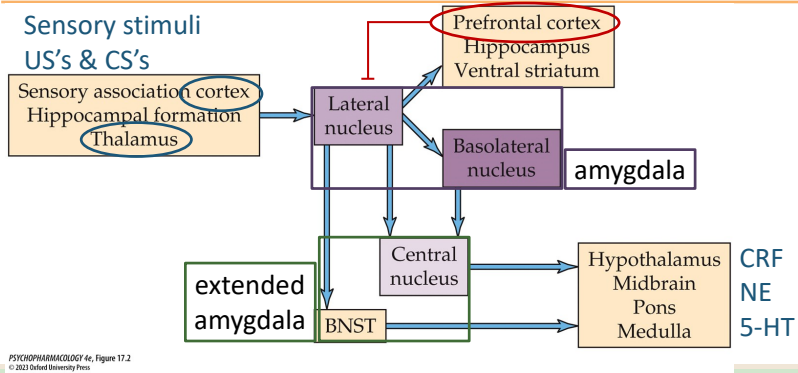
Fear learning is a form of associative (Pavlovian) learning

Hypothalamus & Brainstem

- autonomic and endocrine systems
- freezing, startle

The amygdala is critical for fear learning

Sensory stimuli US's & CS's



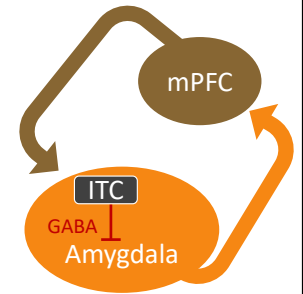
PSYCHOPHARMACOLOGY 4e, Figure 17.2
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The medial prefrontal cortex (mPFC) controls amygdala activity

The mPFC integrates and evaluates several kinds of information

- sensory, context, emotional salience, motivation
- cognition; subjective experience (feelings)

mPFC controls amygdala processing, largely through modulating GABAergic neurotransmission

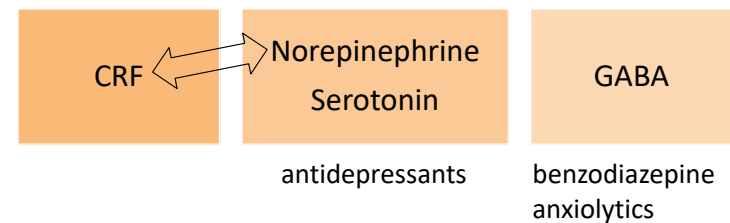


The mPFC provides “top-down” inhibition over fear-related amygdala output

There are 2 primary reasons the mPFC inhibits fear responses

- Escape from the danger is successful
- The threatening stimulus is re-coded as “safe”

Many transmitter systems are implicated in anxiety



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Environmental stressors activate the (hypothalamic-pituitary-adrenal, HPA) stress axis

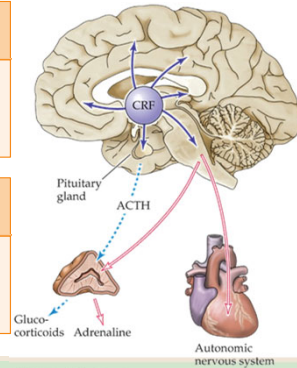
CRF acts as both a hormone and a neurotransmitter

Hormone

- endocrine control
- cortisol release

Neurotransmitter

- amygdala (CeA)
- locus coeruleus (LC)

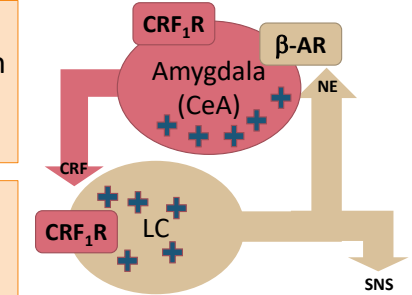


PSYCHOPHARMACOLOGY 4e, Figure 17.4
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CRF neurons in CeA project to LC and activate the adrenergic component of the stress response

Stress increases activity in CRF-releasing neurons from the CeA that drive high-tonic firing in LC neurons.

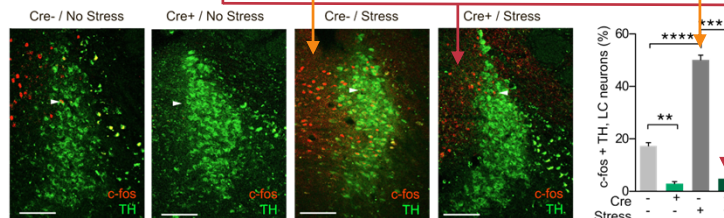
LC activity enhances sympathetic NS activity & further drives the CeA.



Adapted from Koob G. F. and Volkow, N.D. (2010). Neurocircuitry of addiction. *Neuropsychopharmacology*, 35, 217-238. doi: 10.1038/npp.2009.110

Stress and CRF induce high-tonic firing of LC neurons and increase anxiety-like behavior.

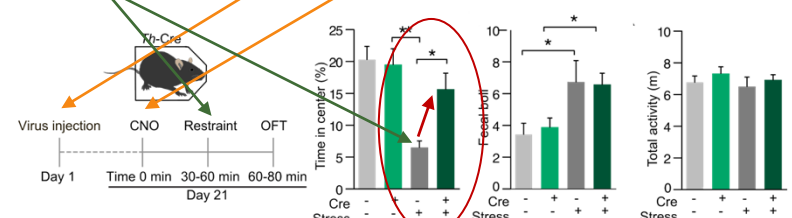
Restraint stress increases neural activity (*c-fos* expression) in the LC that is inhibited by G_{α_i} -coupled DReADD.



McCall, J. G., Al-Hasani, R., Siuda, E.R., Hong, D. Y., Norris, A. J., Ford, C. P. & Bruchas, M. R. (2015). CRH engagement of the locus coeruleus noradrenergic system mediates stress-induced anxiety. *Neuron*, 87, 605-620. doi: 10.1016/j.neuron.2015.07.002

Stress and CRF induce high-tonic firing of LC neurons and increase anxiety-like behavior.

Activating the G_{α_i} -coupled DReADD in TH⁺ LC neurons prior to restraint stress inhibits stress-increased anxiety-like behavior



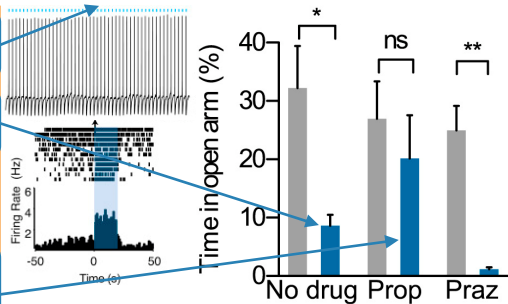
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Stress and CRF induce *high-tonic firing* of LC neurons and increase anxiety-like behavior.

High-tonic firing of LC neurons

increases anxiety-like behavior on the elevated plus maze in rodents that is blocked by β -AR but not α_1 -AR antagonists



McCall, J. G., Al-Hasani, R., Siuda, E.R. Hong, D. Y., Norris, A. J., Ford, C. P. & Bruchas, M. R. (2015). CRH engagement of the locus coeruleus noradrenergic system mediates stress-induced anxiety. *Neuron*, 87, 605-620. doi: 10.1016/j.neuron.2015.07.002

Stress and CRF induce *high-tonic firing* of LC neurons and increase anxiety-like behavior.

Some LC neurons respond to stress-induced CRF with high-tonic firing.

Restraint stress causes LC neurons to increase firing, which is inhibited by $G\alpha_i$ DReADD

- Necessary: Inhibiting the firing with the $G\alpha_i$ DReADD decreased anxiety-like behavior
- Sufficient: Driving the high tonic firing increased anxiety-like behavior, which involved β -AR

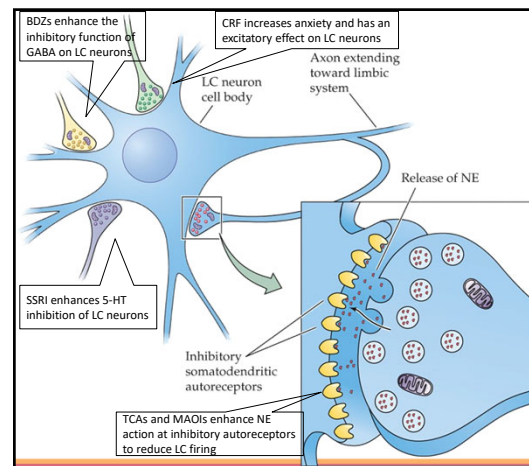
Interim summary

Brain circuits in fear & anxiety

- Amygdala function is modified by mPFC.
- Interactions between the amygdala (threat) and the mPFC (safety) are important in anxiety and stress-related disorders

Transmitter systems implicated in anxiety

- CRF causes NE release; NE causes more CRF release
- CRF transforms LC to high-tonic firing → anxiety



Several medications used to treat anxiety alter LC firing

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NE receptor ligands are used to alleviate physical symptoms of anxiety

prazosin α_1 -AR antagonist

- increases peripheral vasodilation ($\downarrow$ bp)
- reduces insomnia & nightmares in PTSD

propranolol β -AR antagonist

- negative chronotropic & negative inotropic effects on heart
- reduces performance anxiety in social anxiety disorder

Serotonin (5-HT) has a role in anxiety

SSRIs

- have anxiolytic properties
- paroxetine (Paxil), escitalopram (Lexapro)

Mechanism

- not entirely clear
- several different 5-HT receptors with differing distributions

Candidates

- 5-HT_{1A} receptor
- 5-HT transporter gene (SERT)

5-HT_{1A} KO mice exhibit an anxious phenotype

5-HT_{1A} receptors are both somatodendritic (activity-modulating) **autoreceptors** and **postsynaptic receptors**.

autoreceptors

- dorsal & median raphe
- inhibit neuronal firing & 5-HT release
- agonists are anxiolytic

postsynaptic receptors

- many brain area, e.g. hippo & amygdala
- emotional and cognitive aspects
- agonists are anxiogenic

Inability to control stressors enhances 5-HT release, and sensitizes anxiety-like behaviors

Learned helplessness (LH) Inescapable shock

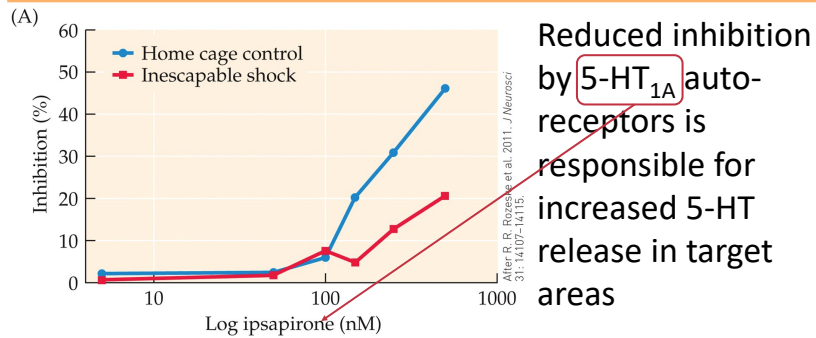
Raphé lesions
prevent behavioral
correlates of LH

Lack of stressor controllability

- inability to control stressor increases 5-HT output (anxiogenic)
- sensitizes postsynaptic targets to 5-HT, and exaggerates anxious behaviors when challenged by future stressors
- due in part to desensitization of 5-HT_{1A} autoreceptors in raphe

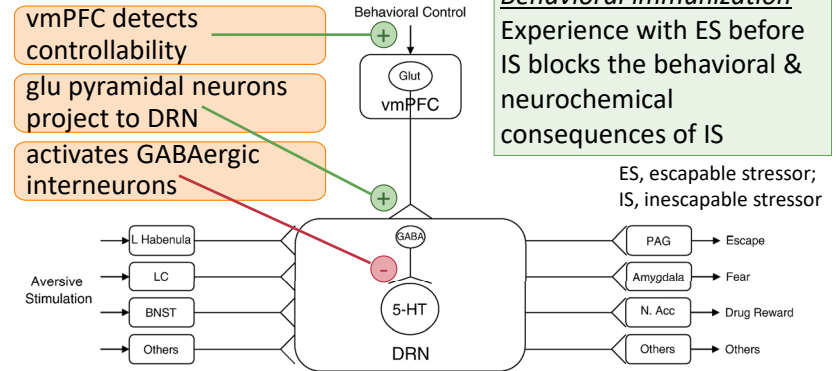
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Inability to control stressors enhances 5-HT release, and sensitizes anxiety-like behaviors



PSYCHOPHARMACOLOGY 4e, Figure 17.9 (Part 1)
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Perceived control over stressors blunts reactions to subsequent stressors.



Maier, S. F. and Watkins, L. R. (2010). Role of the medial prefrontal cortex in coping and resilience. *Brain Research*, 1355, 52-60. doi: 10.1016/j.brainres.2010.08.039

SERT is also implicated in anxiety

Acute SSRI

can elicit anxiety

activation of postsynaptic 5-HT receptors

Chronic SSRI

is anxiolytic

desensitizes 5-HT_{1A} autoreceptors

Genetic variations in SERT affect brain development and anxiety

Short allele

smaller amygdala & sgACC volume

increased amygdala activity

greater symptom severity in anxiety disorders

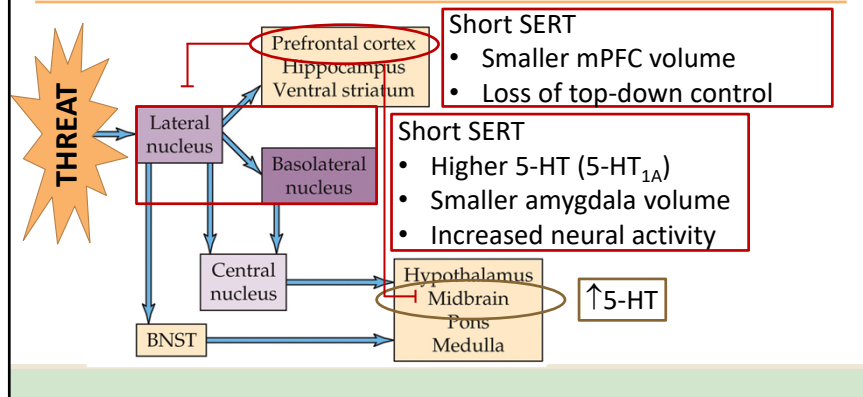
Developmental mechanism:
Early elevated 5-HT

Short allele is low-expressing (less reuptake)

SERT KO or 5-HT_{1A} KO mice are more anxious

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Early life stress confers risk to developing mood and anxiety disorders



Summary so far ...

5-HT can have anxiolytic or anxiogenic effects

- depending on receptor location & development

Both CRF and 5-HT systems are developmentally & epigenetically regulated

- Long-term effects of early adverse events

GABA, NE & 5-HT are targets of anxiolytic medications

Pharmacotherapy for anxiety disorders

Brain circuits in fear & anxiety

Transmitter systems implicated in anxiety

- CRF, NE & 5-HT, GABA

Anxiolytics & the treatment of anxiety

Antidepressants are first-line medications in the treatment of anxiety disorders

Anxiety and depression are related

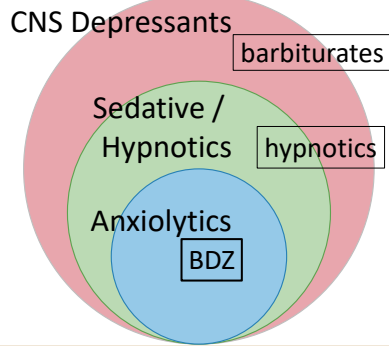
- The occurrence of an anxiety disorder increases the risk of major depression
- People with depression often experience severe anxiety

Antidepressant medications are also anxiolytic

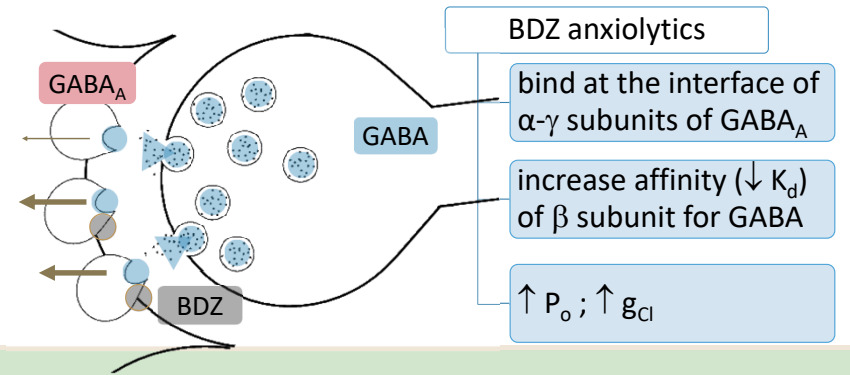
- SSRI, SNRI
- tricyclic antidepressants (TCA)

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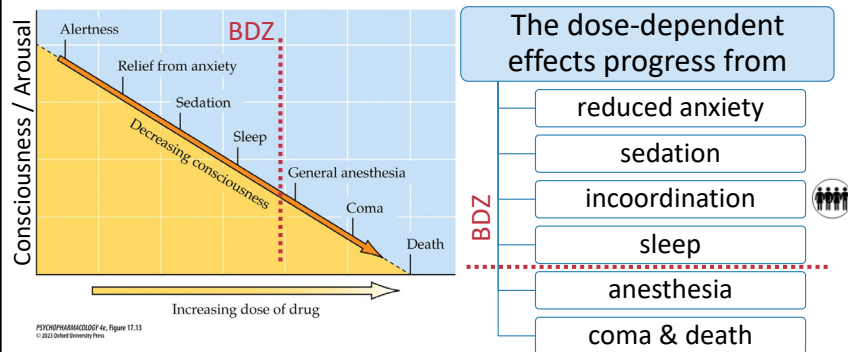
Benzodiazepines are anxiolytics



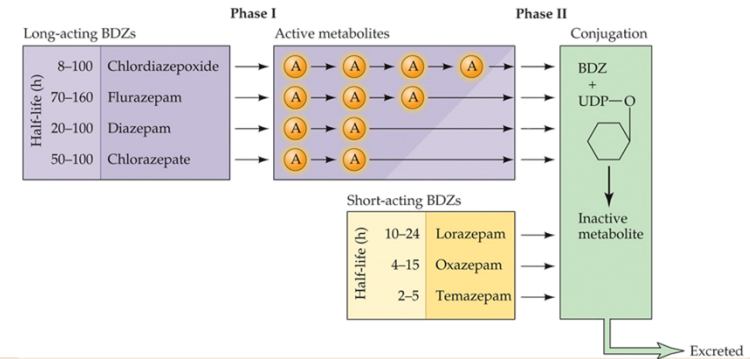
BDZs are positive allosteric modulators of GABA_A receptors



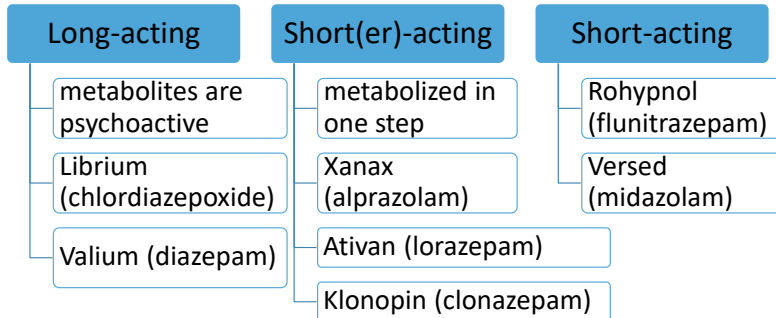
General properties of CNS depressants



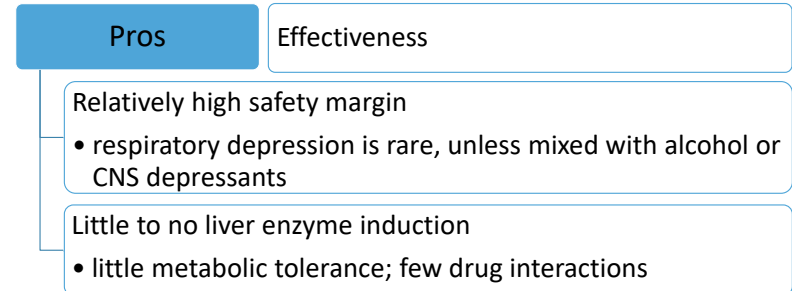
Benzodiazepines are grouped by PK properties



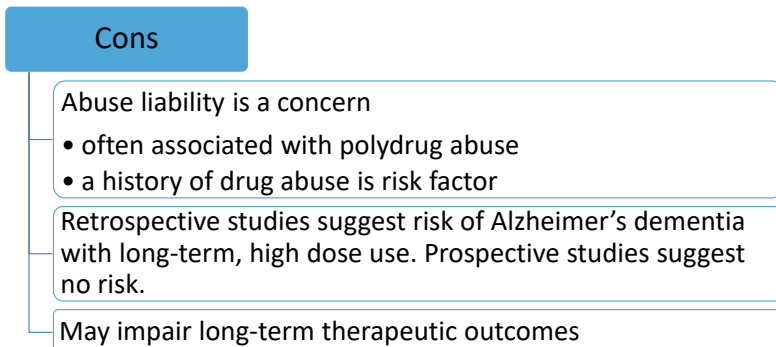
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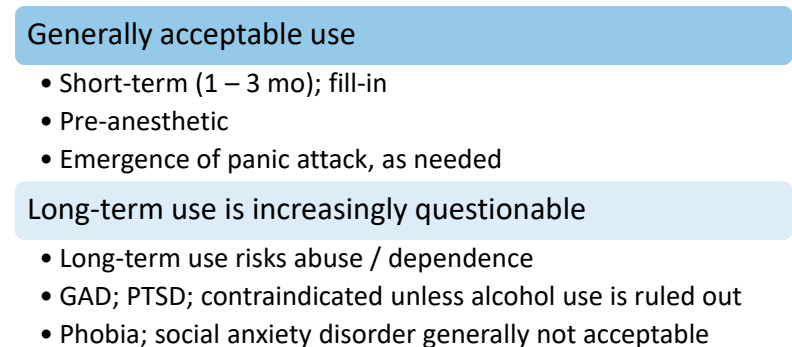
The use of BDZ in the treatment of anxiety disorders



The use of BDZ in the treatment of anxiety disorders



The use of BDZ in the treatment of anxiety disorders



2nd generation anxiolytics have a different psychopharmacological profile

BuSpar (buspirone)

- a non-BDZ anxiolytic; does not act on GABA_A receptors
- buspirone is a partial agonist of 5-HT_{1A} receptors
- anxiolytic effects are slow, follow a “therapeutic lag”

No sed/hypno side-effects; no abuse potential

2nd generation anxiolytics have a different psychopharmacological profile

BuSpar (buspirone)

Useful for cognitive symptoms

- treating comorbid anxiety & depression
- relief of worry, poor concentration; less effective on physical symptoms.

Some anticonvulsants may also be useful

Neurontin (gabapentin) Lyrica (pregabalin)

- block active calcium channels ($\alpha_2\delta$)
- reduces excitatory neurotransmission
- off-label use

Side-effects generally in the cognitive domain

- mental fogging; cognitive slowing
- memory deficits
- dizziness
- sleepiness, drowsiness
- weakness / lethargy

Summary: Pharmacological treatment of anxiety

